

Sec. 2.7: Proof by cases

- A proof by cases of a universal statement such as " $\forall x, P(x)$ ", or "If x is an integer, then..." breaks the domain for the variable into different classes and gives a different proof for each class.
- Every value in the domain must be included in at least one class.

Example 1. Prove that for any integer n , $n^2 + n$ is even.

Proof. • **Case 1:** Let n be an integer.

Since n is even, for some integer k .

Then $n^2 + n =$.

Since is an integer, is also an integer.

Hence, .

• **Case 2:** Let n be an integer.

Since n is odd, for some integer k .

Then $n^2 + n =$.

Since is an integer, is also an integer.

Hence, .

□

Common ways to split the cases

(a) Statement whose hypothesis is an "or" statement:

"If $x > 3$ or $x \leq 1$, then...."

"If $x^2 + 2x - 3 \geq 0$, then...."

(b) Statement about integers:

"If x is an integer, then $x^2 + 3x + 1$ is odd."

"For every integer n , $n^2 \geq n$."

(c) Statement about real numbers:

(d) Statement about absolute values:

"Without loss of generality"

Sometimes the proofs for two different cases are so similar, that it is repetitive to include both cases. When this happens, the two cases can be merged into one case.

Example: "If $a = 0$ or $b = 0$, then $ab = 0$."

"For any two integers x and y , if x is even or y is even, then xy is even. "